

Introduction

Scarf-Brouwer-Nash Pipeline

We formalize a classical route from finite combinatorics to game-theoretic existence: Scarf's room-counting argument gives the discrete structure, Brouwer converts limiting colorful simplices into fixed points, and the product-simplex construction carries the fixed-point theorem to mixed strategy spaces. The poster highlights the Lean interfaces that make these transfers modular rather than one large monolithic proof.



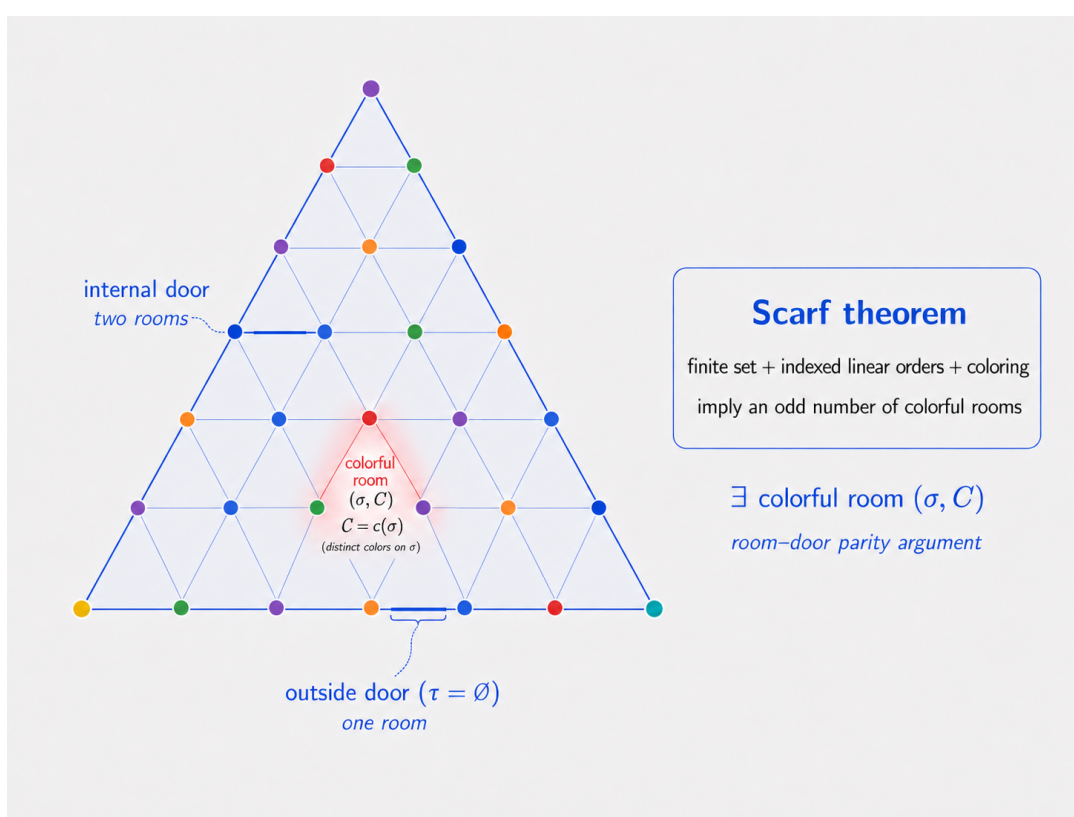
Mathematical Formalization

Scarf's Combinatorial Theorem

Theorem 1. For every coloring $c : T \rightarrow I$, there is at least one colorful room.



Herbert Scarf



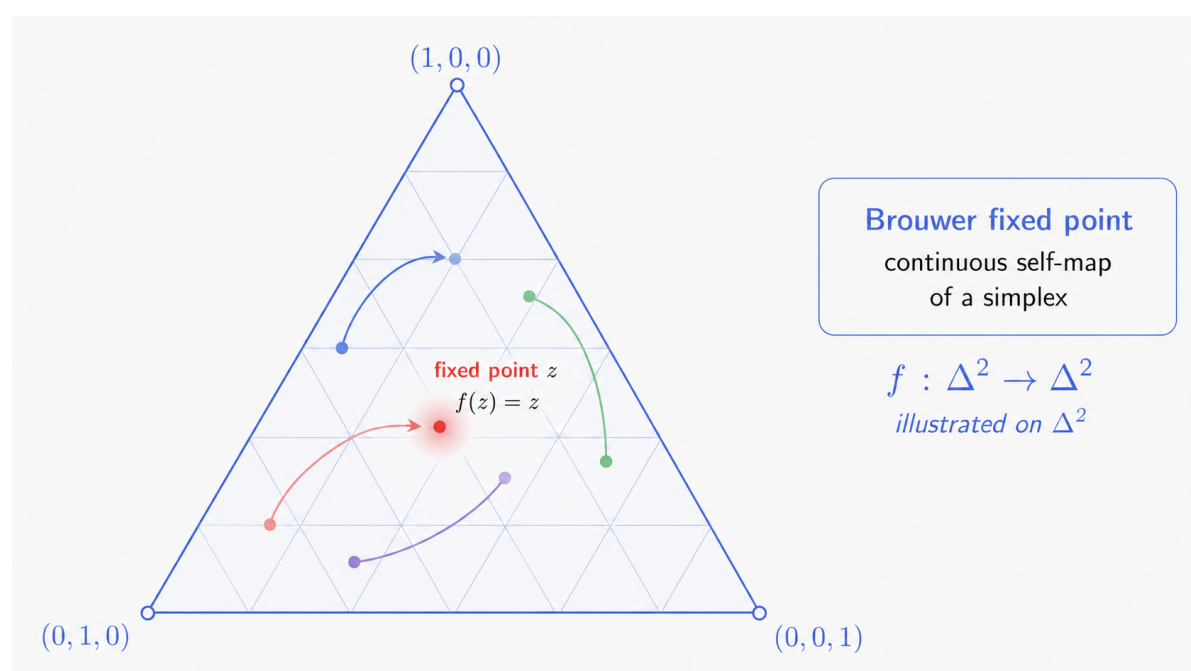
```
abbrev isRoom := isCell σ C ∧ C.card = σ.card
abbrev isDoor := isCell σ C ∧ C.card = σ.card + 1
def isColorful : Prop := isCell σ C ∧ σ.image c = C
abbrev colorful :=
  Finset.filter (fun x => isColorful c x.1 x.2) univ
theorem Scarf : (IST.colorful c).Nonempty := by
  ...
```

Brouwer's Fixed Point Theorem

Theorem 2. Let $\Delta^{n-1} \subset \mathbb{R}^n$ be the standard simplex. Every continuous self-map $f : \Delta^{n-1} \rightarrow \Delta^{n-1}$ has a fixed point z with $f(z) = z$.



L. E. J. Brouwer



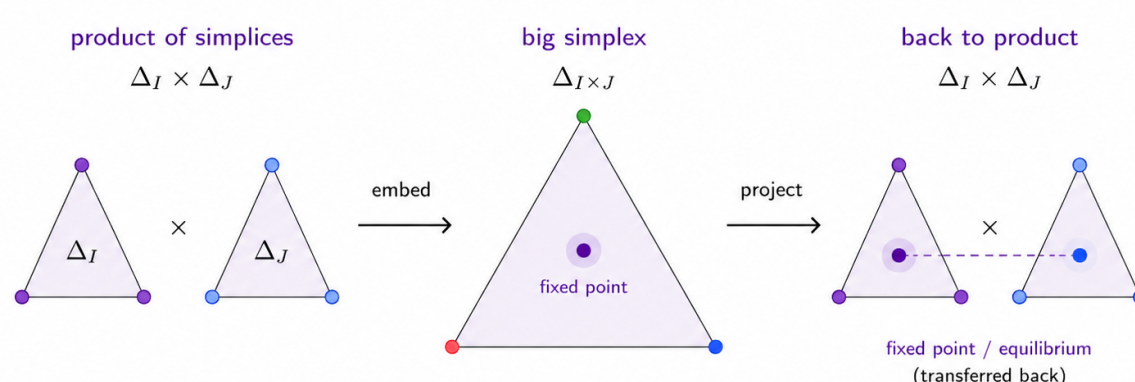
```
abbrev TT := {x : Π i, (c : Fin n) → Fin (i+1) | ∑ i, (x i : ℕ) = 1}
def TTtostdSimplex (x : TT n l) : stdSimplex ℝ (Fin n) := ...
def fcolor (x : TT n l) : Fin n :=
  stdSimplex.pick x (f x)
theorem size_bound_in (σ : Finset (TT n l)) (C : Finset (Fin n)) (h :
  → TT.ILD.isDominant σ C) :
  ∀ x ∈ σ, ∀ y ∈ σ, ∀ i : Fin n, abs ((x i : ℤ) - (y i : ℤ)) < 2 * (n + 1) := by
  ...
theorem Brouwer (hf : Continuous f) : ∃ x, f x = x := by
  ...
```

Brouwer Product Theorem

Theorem 3. For a finite index type I and positive block sizes $card : I \rightarrow \mathbb{N}_{>0}$, every continuous self-map of

$$\prod_{i \in I} \Delta^{card(i)-1}$$

has a fixed point.



- embed_from_product places a product profile in BigSimplex.
- Projection reads blocks back; pushTowardsZ keeps block sums positive.
- project_embed_id is the retraction identity that transfers the fixed point.

Lean bridge. Lift f to BigSimplex, apply Brouwer, then project the fixed point back to the product strategy space.

```
abbrev BigSimplex :=
  stdSimplex ℝ (Fin (total_card card))

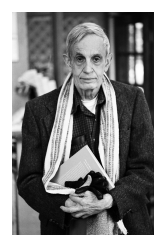
abbrev ProductSimplices :=
  (i : I) → stdSimplex ℝ (Fin (card i))

noncomputable def blockSum (i : I) (x : BigSimplex card) : ℝ :=
  ...
noncomputable def pushTowardsZ (x : BigSimplex card) : BigSimplex card :=
  ...
noncomputable def project_to_product (x : BigSimplex card) : ProductSimplices card :=
  ...
noncomputable def embed_from_product (y : ProductSimplices card) : BigSimplex card :=
  ...
lemma project_embed_id (y : ProductSimplices card) :
  project_to_product card (embed_from_product card y) = y := by
  ...
theorem Brouwer_Product
  (f : ProductSimplices card → ProductSimplices card) (hf : Continuous f) :
  ∃ x : ProductSimplices card, f x = x := by
  ...
```

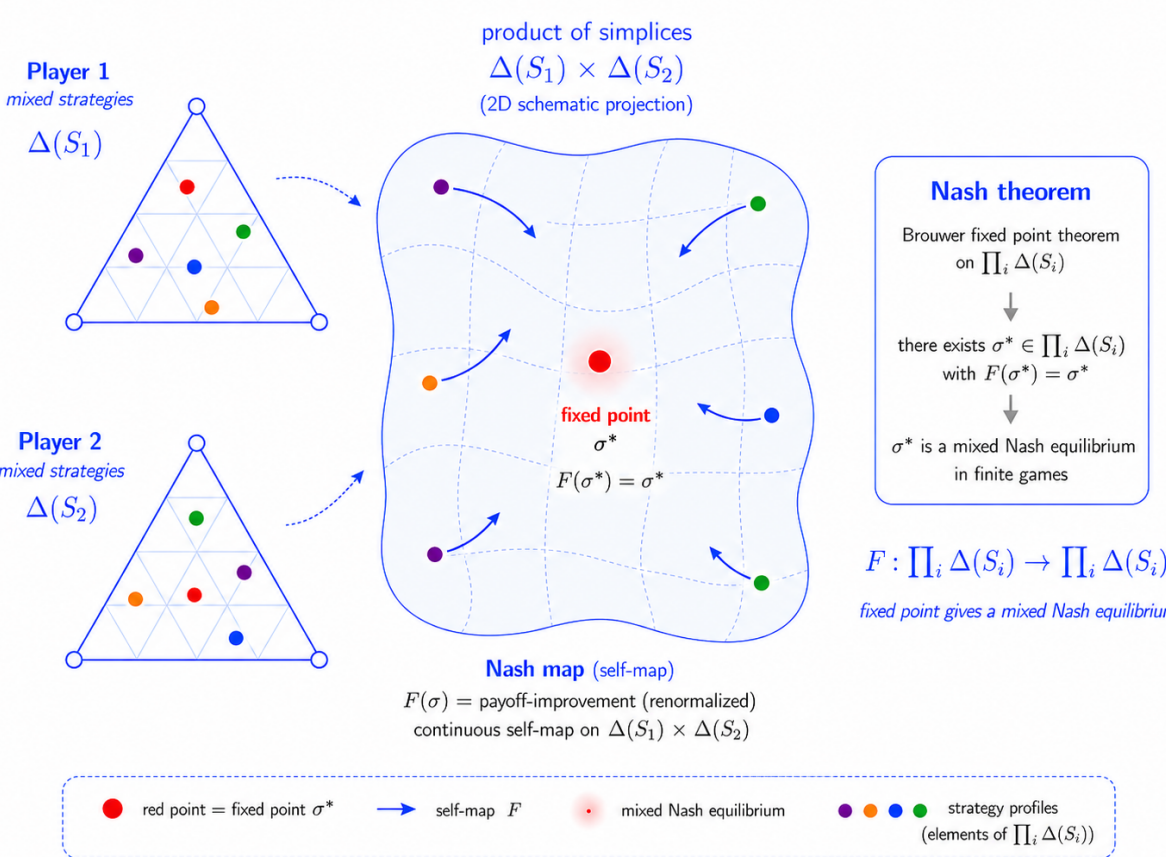
Finite-Game Nash Equilibrium

Theorem 4. Every finite strategic-form game has a mixed Nash equilibrium: there is $\sigma^* \in \prod_{i \in I} \Delta(S_i)$ such that, for every player i and mixed deviation τ_i ,

$$\bar{g}_i(\sigma^*) \geq \bar{g}_i(\sigma_{-i}^*, \tau_i).$$



John Nash



- Mixed profiles are products of player simplices, so the product Brouwer theorem applies directly to this strategy space.
- The Nash map increases weights on profitable pure deviations and then normalizes back into each player's simplex.
- A fixed point of this map forces all positive improvements to vanish, giving the no-profitable-deviation condition.

The Nash proof is therefore a transport argument: continuity gives a fixed point of the normalized improvement map, fixedness rules out profitable pure deviations, and mixed_g_linear lifts those bounds to mixed deviations.

```
structure FinGame extends Game where
  FinI : Fintype I
  FinSS : ∀ i : I, Fintype (SS i)

namespace FinGame
variable {G : FinGame}

abbrev mixedS := (i : G.I) → stdSimplex ℝ (G.SS i)

def mixed_g (i : G.I) (m : Π i, S (G.SS i)) : ℝ :=
  ∑ s : (Π j, G.SS j), (Π j, m j (s j)) * (G.g i s)

noncomputable def nash_map (σ : G.mixedS) : G.mixedS := ...

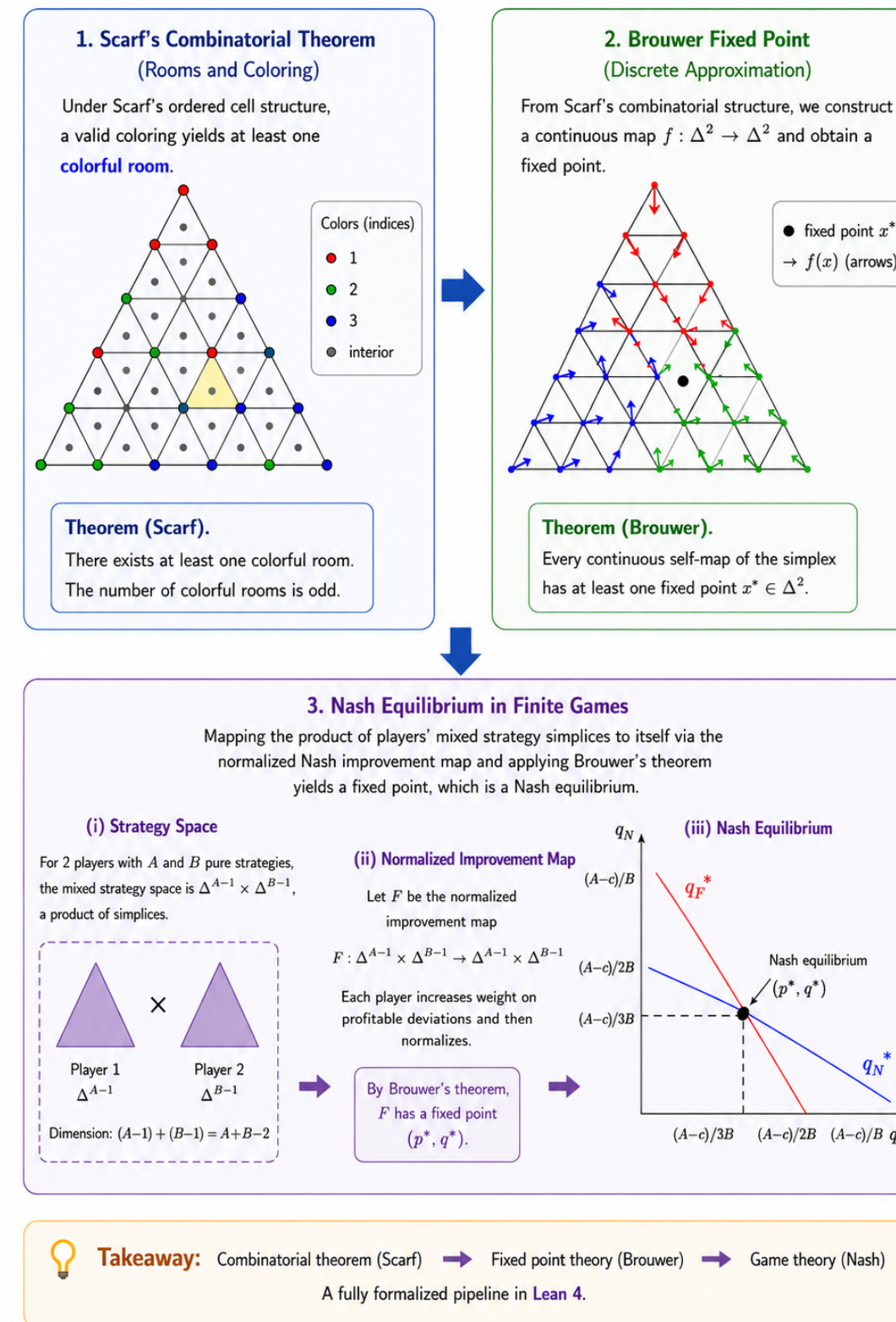
lemma nash_map_cont : Continuous $ nash_map G := by
  ...
def mixedNashEquilibrium {G : FinGame} (x : G.mixedS) :=
  ∀ (i : G.I), ∀ (y : S (G.SS i)),
    G.mixed_g i x ≥ G.mixed_g i (update x i y)
theorem ExistsNashEq : ∃ σ : G.mixedS, mixedNashEquilibrium σ := by {
  ...
```

Lean status. The checked Lean files now prove the four endpoint statements: Scarf for colorful rooms, Brouwer for simplex fixed points, Brouwer_Product for product simplices, and ExistsNashEq for finite games. BrouwerBench turns these proof dependencies into 80 Lean-grounded QA items.

Flowchart

From Scarf to Brouwer to Nash

A Combinatorial Route to Nash Equilibrium in Finite Games



BrouwerBench

Lean-Grounded Evaluation

- 80 proof-structure QA items over Scarf, Brouwer, Product, and Nash.
- Each item stores Lean context, evidence lines, a gold answer, and a 0-2 rubric.
- Best score: gpt-oss:20b, 122/160 (76.2%).
- Main gaps: endpoint bridges, Brouwer limit estimates, and product projection details.

BrouwerBench Results

80 Lean-grounded QA items, scored with a 0-2 rubric

Model	Overall	Pct.	Scarf	Brouwer	Prod.	Nash
gpt-oss:20b	122/160	76.2%	40/48	25/40	29/36	28/36
qwen3:8b	101/160	63.1%	30/48	22/40	25/36	24/36
gemma3:12b	89/160	55.6%	25/48	21/40	24/36	19/36
kimina-prover:7b	58/160	36.2%	14/48	12/40	14/36	18/36

Best overall: gpt-oss:20b, 122/160 (76.2%)

Conclusion and Next Steps

A single Lean development now connects Scarf's finite rooms, Brouwer fixed points, product simplices, and finite-game Nash equilibria through reusable theorem interfaces. The main technical bridge is the product-simplex layer, which turns a one-simplex fixed-point theorem into a tool for mixed strategy spaces. Next, we will streamline the door-room incidence and typed-parity layer, tighten the projection/embedding API, and expand BrouwerBench into theorem-level regression tests for future formalization agents.

Acknowledgements

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References

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